

Correlation Analysis Report

Chronic Disease Burden & Social Determinants Dashboard

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Organization: Global AI
Period: Week 2, Part-Time Internship

Overview

This report presents the correlation analysis completed as part of Week 2. The goal was to quantify how social determinants of health (SDOH) and Social Vulnerability Index (SVI) components relate to county-level chronic disease prevalence across the United States. Pearson correlation coefficients were computed for two outcomes (diabetes prevalence and hypertension prevalence) using data from approximately 2,956 counties. The analysis covers the five-year window from 2019 to 2023 and is summarised across four figures.

Five-Year Trend Stability

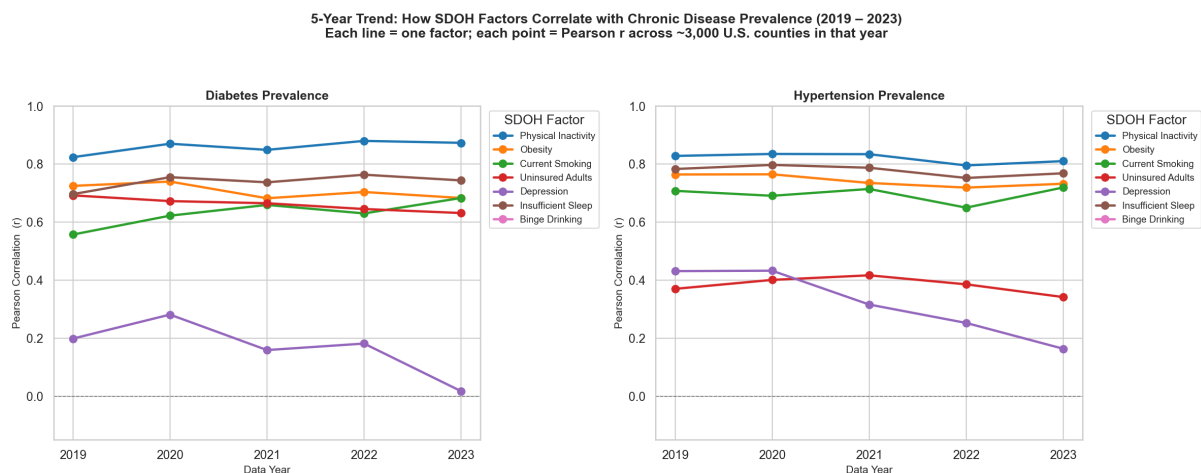


Figure 1: Pearson correlations between seven SDOH factors and chronic disease prevalence, computed annually across ~3,000 U.S. counties from 2019 to 2023.

The first thing that stands out in Figure 1 is how stable most of these relationships are. The lines barely move across five years, which suggests that the structural links between behavioural risk factors and county-level chronic disease burden are not artefacts of any single data year. They are consistent patterns.

Physical inactivity is the dominant factor throughout, holding near $r = 0.82$ – 0.87 for diabetes and $r = 0.81$ – 0.83 for hypertension across the full period. Obesity, current smoking, uninsured adults, and insufficient sleep all cluster in the moderate- to-strong range ($r = 0.60$ – 0.75) with very little variation year to year.

The one line that does move is depression. Its correlation with diabetes prevalence falls steadily from $r = 0.20$ in 2019 to near zero by 2023, and a similar decline appears for hypertension. It is not immediately clear why that is happening. It could reflect changing measurement practices, differences in how

depression is coded in PLACES across releases, or a genuine shift in the county-level relationship. It is worth flagging as something to revisit when the clustering analysis is done.

Change Over Time: 2019 vs. 2023

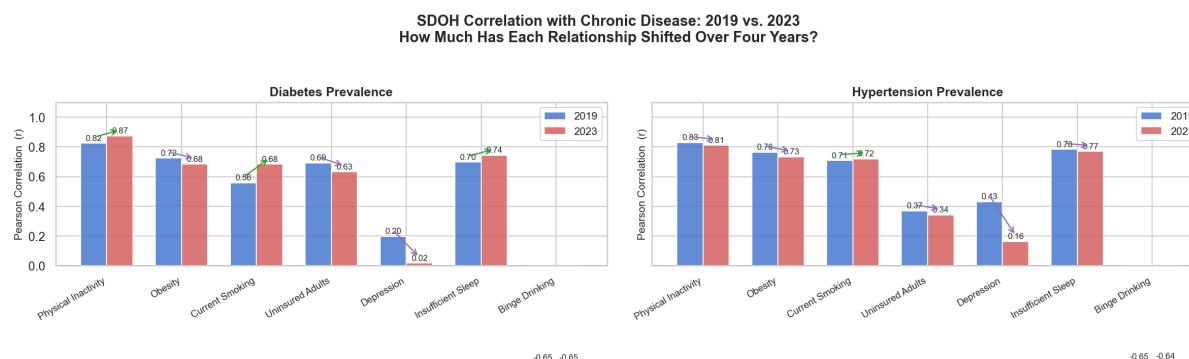


Figure 2: Grouped bar chart comparing SDOH correlations with chronic disease prevalence at the 2019 and 2023 endpoints. Arrows indicate the direction and magnitude of change.

Figure 2 zooms in on the two endpoints to make shifts easier to read. A few patterns are worth noting.

For diabetes, insufficient sleep strengthened from $r = 0.70$ to $r = 0.74$ and current smoking rose from $r = 0.56$ to $r = 0.68$ over the four-year gap. Both of those moves are in the same direction, suggesting that these factors have become more tightly coupled with county-level diabetes burden over time. Obesity moved slightly the other way, from $r = 0.72$ down to $r = 0.68$. Depression's near-zero value in 2023 ($r = 0.02$) is the starkest change in the panel.

Hypertension correlations were more stable across the board. Physical inactivity, obesity, current smoking, and insufficient sleep all held within a few hundredths of their 2019 values. The largest shifts were again in depression ($r = 0.43$ down to $r = 0.16$) and binge drinking, which fell modestly from $r = 0.78$ to $r = 0.77$.

Full Factor Heatmap: 2023 Snapshot

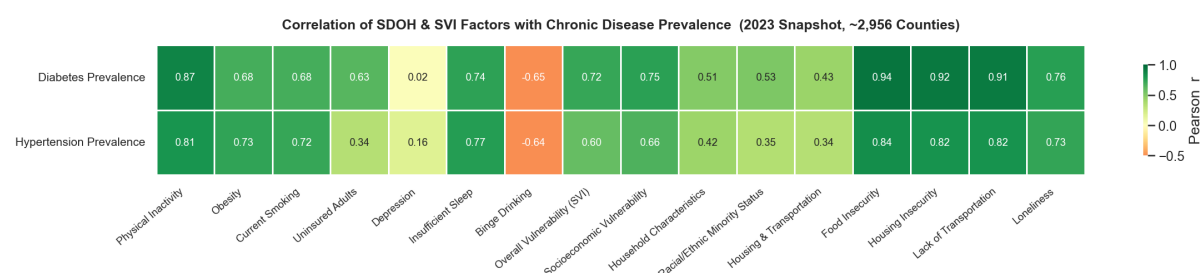


Figure 3: Pearson correlations for all SDOH and SVI factors against diabetes and hypertension prevalence. 2023 cross-sectional snapshot, ~2,956 U.S. counties.

Figure 3 expands the picture by bringing in the SVI components alongside the behavioural SDOH factors. Two things immediately stand out.

First, the SVI-derived measures (food insecurity, housing insecurity, and lack of transportation) are not just correlated with chronic disease; they are among the strongest correlates in the entire feature set, sitting above most of the behavioural factors. That is a meaningful finding for a dashboard project,

because it suggests that any model or visualisation that focuses only on behaviour and ignores material deprivation is missing a large part of the story.

Second, binge drinking is the only factor in the heatmap that goes orange, with a negative correlation of $r = -0.65$ for diabetes and $r = -0.64$ for hypertension. That is counterintuitive and probably not what most people would expect. The most likely explanation is a compositional effect: counties with higher binge drinking rates tend to be whiter, wealthier, and more rural in ways that correlate with lower chronic disease burden through other pathways. It is not a causal story; it is a confounding one. It is worth noting explicitly in the dashboard so it does not get misread.

Ranked Correlates: 2023

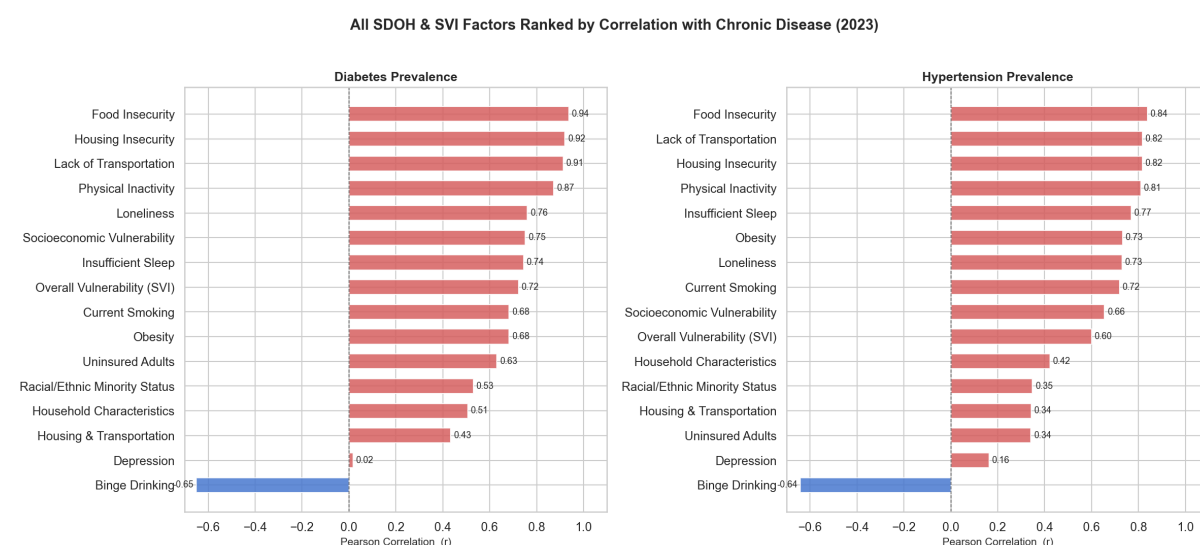


Figure 4: All SDOH and SVI factors ranked by Pearson correlation with diabetes prevalence (left) and hypertension prevalence (right) in 2023. Blue bar indicates negative correlation.

Figure 4 puts everything in order. Food insecurity comes out on top for both outcomes ($r = 0.94$ for diabetes and $r = 0.84$ for hypertension), making it the single strongest county-level predictor in the dataset. Housing insecurity and lack of transportation follow immediately behind, completing a cluster of material deprivation measures at the top of both rankings.

Physical inactivity ($r = 0.87$ / $r = 0.81$) is the highest-ranked behavioural factor and the first non-SVI variable to appear in both lists. After that, the behavioural and SVI factors are more interleaved, with loneliness, socioeconomic vulnerability, and insufficient sleep all sitting in the mid-range.

Depression and binge drinking sit at the bottom. Binge drinking is the only negative correlate, and its position there is consistent with the confounding explanation described above.

Summary of Key Findings

- **Correlation patterns are stable over time.** Most SDOH–disease relationships changed very little between 2019 and 2023, which gives some confidence that the 2023 snapshot is representative rather than a one-year anomaly.
- **Material deprivation dominates the rankings.** Food insecurity, housing insecurity, and lack of transportation are the strongest county-level predictors of both diabetes and hypertension, stronger than any single behavioural factor.

- **Physical inactivity is the top behavioural predictor.** It holds consistently near $r = 0.85$ – 0.87 for diabetes and $r = 0.81$ – 0.83 for hypertension across all years.
- **Depression's correlation has declined sharply.** The drop from $r \approx 0.20$ (2019) to near zero (2023) for diabetes is the largest single shift in the dataset and warrants further investigation.
- **Binge drinking is a negative correlate.** This is most likely driven by confounding rather than a protective effect, and should be labelled clearly in any dashboard output.